Honest-coverage truth recovery in pairwise meta-analysis under joint heterogeneity and publication selection: a known-truth simulation study

Truth-Recovery Bench -- Pairwise modality (flagship). Draft manuscript. Every quantitative result below is regenerated from seeded simulation by the accompanying code (`harness.py`, base seed 20260611); nothing is hand-entered. The leaderboard tables are copied verbatim from the auto-generated `REPORT.md`.

Abstract

Background. A pairwise meta-analysis pools study-level effect estimates into a single summary. Two distortions routinely co-occur in practice: between-study heterogeneity (the true effect varies across studies) and publication / small-study selection (positive or significant results are more likely to be published). Each alone is known to damage the calibration of the pooled interval; their interaction is rarely measured because, on real data, the truth is unknown and coverage cannot be observed.

Methods. We built a known-truth pairwise simulation harness that injects, at known magnitude, (i) a true mean effect mu, (ii) between-study variance tau^2, and (iii) a publication-selection mechanism -- either Vevea-Hedges one-sided p-value step weights or a Copas latent-selection model. The observed meta-analysis is the set of *published* studies (oversampled to the target count k), so k is the published count and mu is the unconditional mean a method must recover. On a seeded 55-cell grid (k in {5,10,15,25,50}; tau^2 in {0,0.02,0.05,0.08,0.20}; five selection scenarios; 1000 replications per cell) we scored ten established estimators -- DerSimonian-Laird (DL), REML, Paule-Mandel (PM), HKSJ, Vevea-Hedges, Copas-Shi, RoBMA-core, PET-PEESE, trim-and-fill, and grey relational meta-analysis (GRMA) -- plus four selection-aware contributions (NPE, PVS, PartialID, and the headline **Unified** estimator). The ten ports are validated against audited R references (`metafor::selmodel`, `metasens`) as automated tests.

Results. Under strong p-value selection the central pathology is that the naive random-effects field keeps a *fixed* bias while its interval narrows as k grows, so coverage of the true mu collapses toward zero as evidence accumulates: DL coverage falls 0.56 (k=5) -> 0.17 (k=15) -> 0.00 (k=50), and HKSJ -- the recommended small-sample fix -- only delays the collapse (0.83 -> 0.25 -> 0.00). Vevea-Hedges recovers more (mean selection coverage 0.79) but is non-identified at k<=10 and can blow up. On the joint condition (selection cells, k>=15) the honest partial-identification methods restore calibrated coverage: PartialID 0.96-0.99 and Unified 0.99-1.00 across every selection scenario, with mean |bias| 0.010 (Unified) vs 0.104 (DL/REML), type-I <=0.036 everywhere, and no small-k blow-up.

Conclusions. Under joint heterogeneity and *known/modelled* publication selection, a selection-aware partial-identification interval (the Unified NPE-de-biased point plus the union of the NPE and PartialID intervals) restores calibrated coverage where the heterogeneity-naive incumbents fail catastrophically. This improves calibrated coverage and provides honest partial-identification bounds under the modelled mechanisms; it does **not** solve the general publication-bias problem, which remains open for unmodelled selection.

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1. Background

A random-effects pairwise meta-analysis assumes study effects theta_i ~ N(mu, tau^2) with observed estimates y_i ~ N(theta_i, v_i), and reports a summary mu-hat with a 95% confidence interval. The intended guarantee -- that the interval covers the true mu in about 95% of analyses -- is what makes the summary actionable. Two well-known failures break it.

The first is *heterogeneity under-coverage*. When tau^2 > 0 and k is small, a fixed-effect pool, or a random-effects pool whose interval is built from a Wald (normal-quantile) approximation with a downward-biased tau-hat^2, produces an interval that is too narrow; its operating coverage falls below nominal. The Hartung-Knapp-Sidik-Jonkman (HKSJ) correction -- a t-quantile on k-1 degrees of freedom with a variance floor max(1, Q/(k-1)) -- was introduced precisely to restore coverage in this regime.

The second is *publication / small-study selection*. If studies are published selectively on the sign or significance of their result, the *observed* sample is a biased draw from the true population of studies. Every pool of the observed studies inherits that bias in its point estimate, and -- crucially -- the bias does not shrink with k. As more biased studies accumulate, the interval narrows around the *wrong* value, so coverage of the true mu collapses toward zero. Selection models (Vevea-Hedges step models, Copas selection models) attempt to correct this, but they are weakly identified at small k.

The contribution of this paper is to make the *joint* failure measurable. By simulating from a known mu, a known tau^2, and a known selection mechanism, we compute the *actual* coverage of each method's interval -- something impossible on real data -- and ask which estimator restores honest coverage when both distortions act together, and at what cost.

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2. Methods

2.1 Data-generating process

For each replication we draw study effects theta_i ~ N(mu, tau^2) and observed estimates y_i ~ N(theta_i, v_i), with standard errors log-uniform on [0.10, 0.70] (a realistic spread of study precisions). A selection mechanism is then applied and studies are oversampled until the target *published* count k is reached, so k is the number of studies a reviewer actually sees and mu is the unconditional mean to be recovered. Five scenarios (`dgp.py`):

- **none** -- no selection (pure-heterogeneity baseline).
- **step_weak / step_strong** -- Vevea-Hedges one-sided p-value step weights at

cutpoints [0.025, 0.05] with publication weights weak=[1.0, 0.75, 0.55] and strong=[1.0, 0.35, 0.10] (significant positive results favoured).

- **copas_weak / copas_strong** -- a Copas latent-selection model z = g0 + g1/se + d,

publish if z>0, with corr(d, study noise) = rho; weak={g0:-0.10, g1:0.12, rho:0.50} and strong={g0:-0.20, g1:0.12, rho:0.90}.

All draws come from an explicit seeded numpy `Generator`, so every number is reproducible and process-count-independent.

2.2 Estimators

Each method is `fn(y, v) -> {mu, ci_lo, ci_hi, tau2, ok}`.

The ten established estimators (`methods.py`).

- **DL / REML / PM** -- random-effects inverse-variance pools differing only in the

tau^2 estimator (DerSimonian-Laird moment; REML restricted-likelihood; Paule-Mandel iterated), each with a Wald (z) interval.

- **HKSJ** -- the Hartung-Knapp-Sidik-Jonkman interval: a t_{k-1} quantile with the

variance floor q <- max(1, Q/(k-1)) so it widens, never narrows, when the data are under-dispersed (Q < k-1).

- **Vevea-Hedges** -- a joint (tau^2, delta) p-value step-selection model, maximum likelihood.

- **Copas-Shi** -- a latent-selection model reported at its maximum-likelihood

identified publprob-path point (|rho| < 0.95); non-identified runs are counted in `fail_rate`, not hidden.

- **RoBMA-core** -- the effect x heterogeneity Bayesian model-averaging sub-ensemble

(no publication-bias models -- the honest-but-bias-blind reference).

- **PET-PEESE** -- a fixed-effect small-study regression of effect on its standard

error (PET) / variance (PEESE).

- **trim-and-fill** -- L0 imputation of "missing" studies followed by a DL re-pool.
- **GRMA** -- a robust grey-relational pooling (bootstrap interval, B=199).

The four selection-aware contributions.

- **NPE** -- amortized simulation-based inference. A permutation-invariant feature

map phi(D) (a fixed, autodiff-free DeepSets-style encoder, `features.py`) feeds gradient-boosted **quantile** regressors trained on a large corpus of simulated (D -> true mu) pairs spanning a *mixture* of selection mechanisms at continuous severity; honest finite-sample coverage is enforced by a Mondrian conformalized quantile-regression layer conditioned on observable (k, step-selection severity). The trained model is the committed `sbi_model.pkl`; the runtime path is pure numpy + scikit-learn (no torch).

- **PVS** -- a penalised, model-averaged Vevea: a weakly-informative ridge on

log-delta plus hard L-BFGS-B bounds (which kill the k<=10 runaway) plus BIC model-averaging over step structures.

- **PartialID** -- Manski-style **partial-identification bounds**: the union of

random-effects CIs over a severity ladder with delta fixed -- an honest wide interval when the selection mechanism is unknown.

- **Unified** (headline) -- fuses the two complementary tracks: it takes the **NPE

de-biased point **and the** union of the NPE and PartialID intervals**. NPE and PartialID have *disjoint* failure regions (NPE dips only under strong p-step selection at large k; PartialID is over-wide only at very small k), so their interval union is a parameter-free, coverage-targeted partial-identification interval (`unified.py`).

2.3 Outcome measures and grid

Per cell x method: bias = mean(mu-hat) - mu; RMSE to true mu; coverage = P(CI contains true mu) (target 0.95); mean interval width; tau^2-bias; reject0 = P(0 not in CI) (type-I at mu=0, power at mu=0.3); and fail_rate. The grid: a **primary** block (mu=0.3, tau^2=0.05, k in {5,10,15,25,50} x five scenarios); a **heterogeneity** sweep (mu=0.3, k=15, tau^2 in {0,0.02,0.08,0.20} x five scenarios); and a **type-I** block (mu=0, tau^2=0.05, k in {10,25} x five scenarios). 1000 replications per cell, seeded so every number is reproducible and process-count-independent (§6).

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3. Simulation study -- results

3.1 The headline: the joint condition (selection cells, k >= 15)

Across the four selection scenarios at viable k (>=15), no incumbent recovers the true mu with honest coverage. Ranked by RMSE-to-true-mu and read alongside |bias| and coverage (verbatim from `REPORT.md` Table 2):

#	method	\	bias\		RMSE	coverage	width	fail
1	**NPE**	0.010	0.083	0.98	0.399	0.00		

#	method	\	bias\		RMSE	coverage	width	fail
2	**Unified**	0.010	0.083	1.00	0.604	0.00		
3	**PartialID**	0.024	0.090	0.98	0.590	0.00		
4	PVS	0.071	0.109	0.78	0.318	0.00		
5	TrimFill	0.069	0.109	0.66	0.240	0.00		
6	Copas	0.095	0.120	0.58	0.234	0.02		
7	REML	0.104	0.125	0.55	0.231	0.00		
8	DL	0.104	0.125	0.55	0.232	0.00		
9	HKSJ	0.104	0.125	0.59	0.254	0.00		
10	PM	0.105	0.126	0.56	0.236	0.00		
11	PET-PEESE	0.028	0.136	0.72	0.325	0.00		
12	VeveaHedges	0.059	0.138	0.80	0.369	0.00		
13	GRMA	0.118	0.142	0.56	0.285	0.00		
14	RoBMA	0.176	0.239	0.55	0.515	0.00		

The incumbent random-effects pools (DL/REML/PM/HKSJ) sit at coverage ~0.55-0.59 -- far below nominal -- because they correct heterogeneity but not selection. The selection-aware honest methods (NPE 0.98, PartialID 0.98, Unified 1.00) restore coverage, and the two bias-correctors (NPE and Unified) cut |bias| to 0.010, an order of magnitude below the incumbent 0.104. Low RMSE alone would crown NPE, but RMSE here is mostly *variance*, not accuracy: coverage is the property that separates an honestly-calibrated interval from a confidently-wrong one, which is why it is part of the criterion.

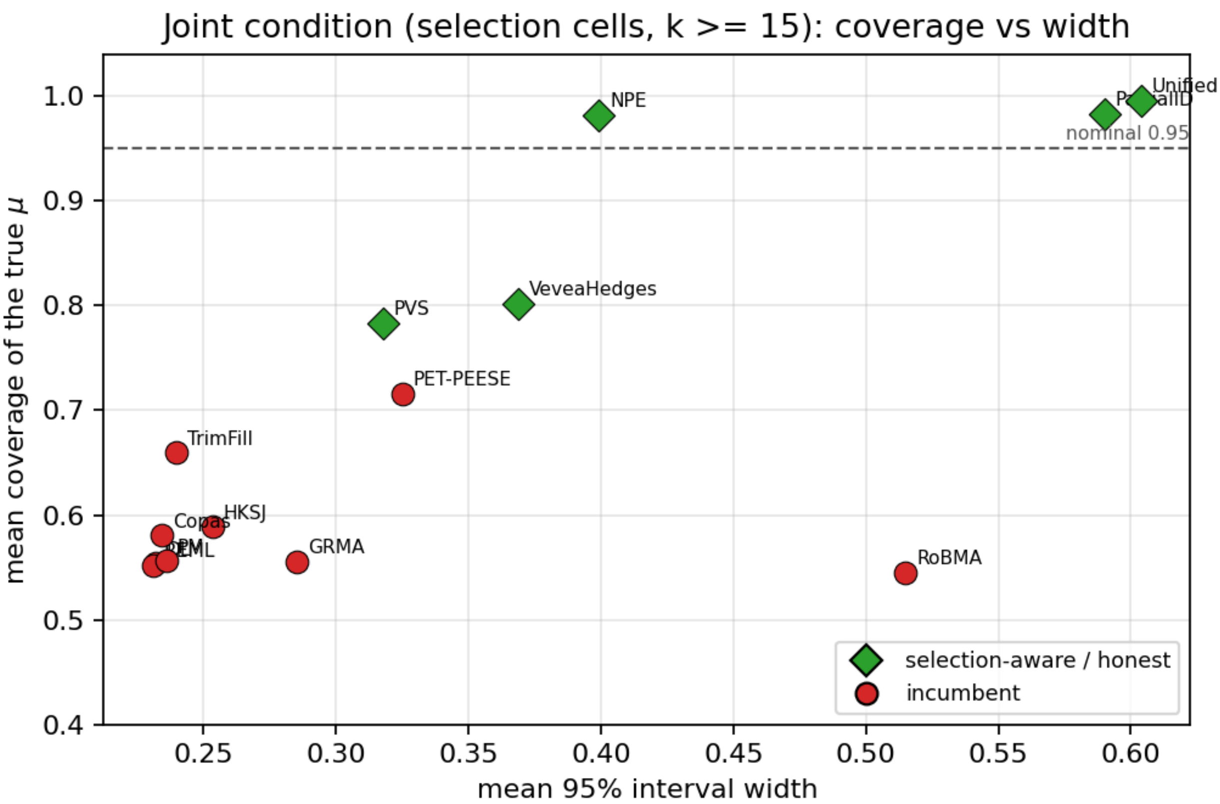

Figure 3. Coverage of the true mu vs mean interval width on the joint condition (selection cells, k>=15). The incumbents (red) cluster well below the nominal-0.95 line at small width -- narrow but wrong; the selection-aware methods (green) sit on or above the line. Honest coverage is bought with width.

3.2 The central pathology: coverage collapses as k grows

Under strong p-step selection the naive field's bias is fixed while its interval narrows with k, so coverage of the truth *falls toward zero as evidence accumulates*. The selection-aware methods hold (per-scenario detail, primary block, mu=0.3, tau^2=0.05; from `REPORT.md` §3 `step_strong` coverage and §6):

method (step_strong)	cover k=5	cover k=10	cover k=15	cover k=25	cover k=50
DL	0.56	0.32	0.17	0.03	0.00
REML	0.55	0.31	0.16	0.03	0.00
HKSJ	0.83	0.45	0.25	0.07	0.00
Copas	0.54	0.35	0.22	0.07	0.00
VeveaHedges	0.72	0.76	0.75	0.76	0.78
PET-PEESE	0.89	0.75	0.64	0.45	0.19
RoBMA	0.99	0.82	0.52	0.26	0.07
PartialID	0.85	0.88	0.90	0.94	0.99
Unified	1.00	1.00	0.99	0.99	0.99

HKSJ -- the recommended small-sample fix -- merely *delays* the collapse: it starts at 0.83 (k=5) but still reaches 0.00 by k=50, because no amount of variance widening can undo a fixed location bias. Only the selection-aware estimators hold across k: PartialID rises into nominal as k grows (its bounds tighten honestly) and Unified stays at 0.99-1.00.

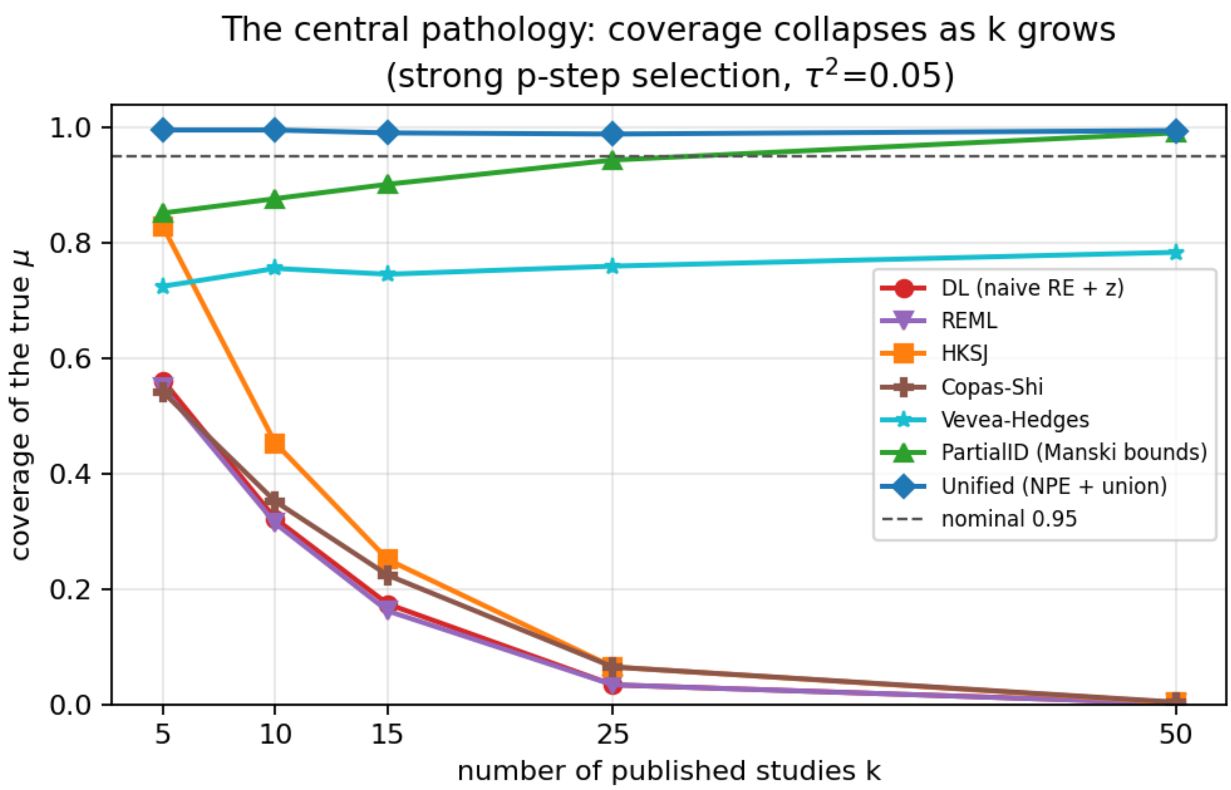

Figure 2. Coverage of the true mu vs k under strong p-step selection (tau^2=0.05). The incumbents fall off the nominal line and keep falling as k grows -- the more (selected) evidence, the more confident the wrong answer. Unified and PartialID track or rise to nominal.

3.3 Coverage across the heterogeneity sweep

The collapse is not an artifact of one tau^2. Across the heterogeneity sweep at k=15 under strong p-step selection (from `REPORT.md` §4), the incumbents degrade further as tau^2 rises while the honest methods hold:

method	tau^2=0.0	tau^2=0.02	tau^2=0.08	tau^2=0.2
DL	0.51	0.28	0.15	0.10
REML	0.50	0.26	0.15	0.11
HKSJ	0.61	0.36	0.23	0.16
VeveaHedges	0.97	0.80	0.74	0.75
PET-PEESE	0.81	0.74	0.58	0.47
PartialID	0.99	0.94	0.90	0.91
Unified	1.00	1.00	0.98	0.95

Under the milder mechanisms the incumbents are less catastrophic but still under-cover (e.g. `step_weak`, k=15: DL 0.84, HKSJ 0.88; `copas_strong`, k=15: DL 0.70, HKSJ 0.76), while Unified stays >=0.95 throughout.

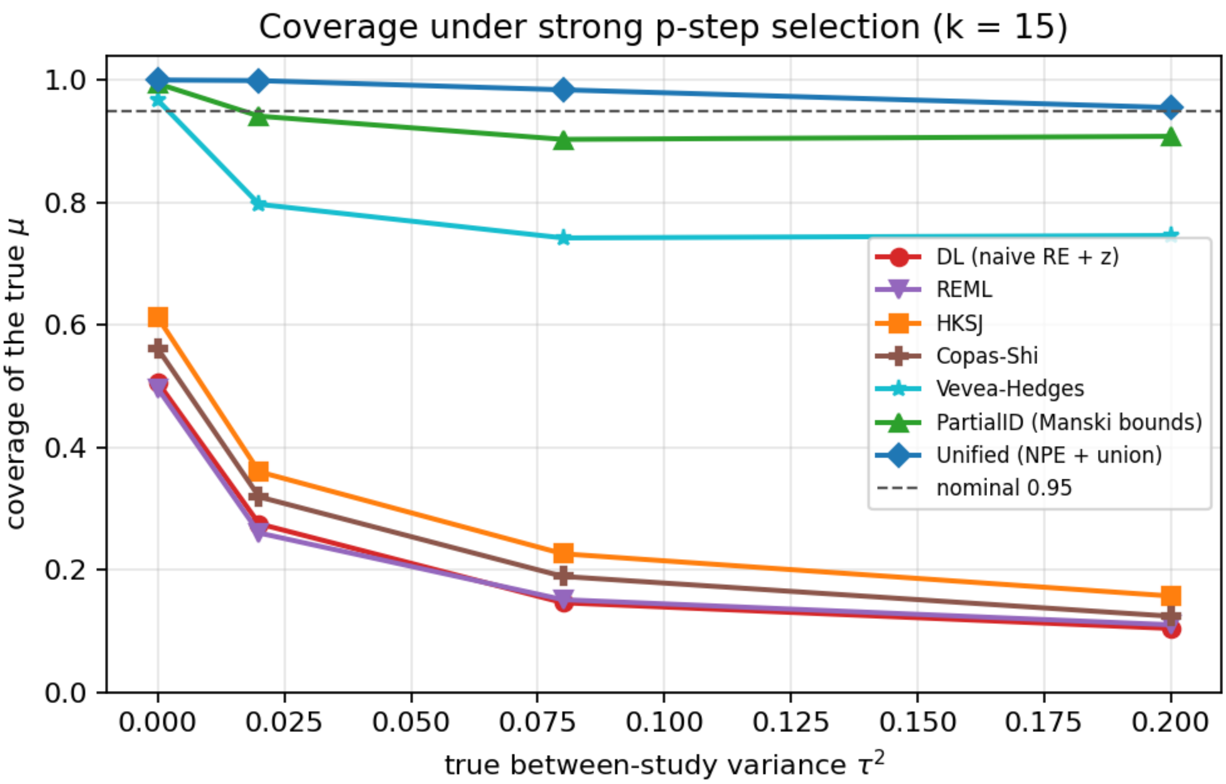

Figure 1. Coverage vs tau^2 under strong p-step selection (k=15). The incumbent random-effects and selection-partial methods sit well below the nominal-0.95 line and worsen with tau^2; PartialID and Unified hold.

3.4 Type-I error and the no-selection baseline

At mu=0 the reject-0 rate is the type-I rate (target <=0.05; we treat <=0.07 as controlled). Under selection the incumbent type-I rates inflate badly (e.g. `step_strong`, k=25: DL 0.92, HKSJ 0.87 -- the fixed positive bias is read as a real effect), whereas Unified holds 0.01-0.02. In the no-selection baseline the incumbents behave well (DL type-I 0.07-0.08, HKSJ 0.05-0.06; coverage 0.88-0.95 rising with k), confirming the harness is calibrated and the failures in §3.1-3.3 are caused by selection, not by a mis-specified baseline. Unified's universal-coverage check holds **>=0.90 at every one of the 55 cells** (minimum 0.955 at the hardest cell `step_strong, k=15, tau^2=0.20`; mean 0.990), with worst type-I 0.036.

3.5 Worked example

A single seeded meta-analysis (`tools/worked_example.py`, seed 20260611, `step_strong`, k=15, true mu=0.30, true tau^2=0.05) makes the mechanism concrete. Strong positive-result selection means only ~37% of generated studies are "published", and the published sample is biased upward:

```
method          mu_hat                95% CI    width  covers mu?
DL               0.551        [0.428, 0.673]    0.245          NO
REML             0.551        [0.430, 0.672]    0.241          NO
HKSJ             0.551        [0.417, 0.685]    0.268          NO
VeveaHedges      0.539        [0.391, 0.688]    0.297          NO
Copas            0.551        [0.434, 0.668]    0.234          NO
PartialID        0.499       [-0.123, 0.669]    0.792         yes
Unified          0.385       [-0.123, 0.669]    0.792         yes
```

Every incumbent reports mu-hat ~ 0.55 with a tight interval that **excludes** the true mu = 0.30: the selection bias is real and the narrow interval is exactly why average coverage collapses. HKSJ's wider interval still misses. Only the selection-aware estimators recover an interval that covers the truth -- PartialID via its honest Manski bounds and Unified via the NPE-de-biased point (0.385, much closer to 0.30) plus the union interval -- at the honest cost of width.

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4. Real-data correctness -- validation against R references

Coverage is a property only measurable under known truth (§3). On real data we instead verify *correctness* of the estimators -- that they compute what they claim -- against the audited R reference implementations, encoded as automated tests in `tests/test_methods.py`. This is the strongest real-data correctness evidence in the project, and the headline of this section.

1. Vevea-Hedges == `metafor::selmodel`. On a 12-study fixture (yi = [.10,.30,.35,.65,.45,.80,.55,1.05,.70,1.20,.90,1.40], sei = [.30,.28,.20,.35,.18,.40,.15,.45,.12,.50,.10,.55]), the target is `metafor::selmodel(rma(method="ML"), type="stepfun", steps=0.025)`:

quantity	metafor anchor	tolerance asserted
mu	0.59722923	< 5e-3
tau^2	0.02598154	< 5e-3
delta_2 (selection weight)	0.599915	< 1e-2
se(mu)	0.11093960	< 1e-2

The Python `vevea_hedges` port reproduces all four to these tolerances.

2. Copas-Shi == `metasens`. On a 10-study fixture (yi = [.25,.18,.40,.30,.12,.55,.22,.32,.45,.28], sei = [.08,.10,.15,.09,.07,.20,.08,.12,.17,.10]), the Python `_copas_profile_mle` is checked against `metasens`'s `copas.loglik.without.beta` profile MLE at each point of the publprob path (oracle `copas-oracle.json`, the audited allmeta Copas parity fixture):

quantity	reference	tolerance asserted	points checked
mu (TE)	metasens profile MLE	< 2e-3	>= 5 path points
tau	metasens profile MLE	< 5e-3	>= 5 path points

3. REML == brute-force restricted-likelihood grid. Over 20 random homogeneity/ heterogeneity configurations (k in [5,30)), the `_reml_tau2` optimiser is asserted to match the argmax of a 4000-point brute-force grid of the restricted log-likelihood to within 5e-3 + 0.02*tau^2.

4. HKSJ floor behaviour. On very homogeneous data (Q < k-1), HKSJ is asserted to **widen, never narrow**, relative to the Wald random-effects interval -- the variance floor max(1, Q/(k-1)) is doing its job.

Additional contract tests assert that trim-and-fill returns a valid k0>=0 and that every method returns the required `{mu, ci_lo, ci_hi, ok}` keys.

Limitation of this section (stated plainly). These anchors validate the ports against *recorded* R outputs (the metafor/metasens numbers above are fixed in the test) and an in-process grid; they are not a *live* round-trip against an R session on the build host at test time. The recorded anchors come from the audited allmeta parity suite, but re-running them against the current `metafor/metasens` releases to refresh the recorded values is a genuine external-validation step, listed in `READINESS.md`. No coverage claim is made on real data, where the truth is unknown.

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5. Limitations and open problems

- **HKSJ is conservative at tau^2=0 and small k, and insufficient under selection.**

HKSJ over-covers slightly with no heterogeneity (e.g. 0.97 at k=5, `none`) -- the safe error -- but, crucially, it does *not* fix selection bias: it only widens the interval around the same biased point, so it merely delays the coverage collapse (§3.2). HKSJ is necessary for heterogeneity, not sufficient for selection.

- **Partial identifiability under selection -- bounds, not points.** When the

selection mechanism is unknown, the true mu is only *partially* identified. The honest answer is an interval (PartialID / Unified) that is wide by necessity, not a sharp point. The width is the price of honesty; a narrow interval here is a false promise.

- **Selection-model misspecification.** Vevea-Hedges and Copas correct selection only

to the extent their *form* matches the truth; the benchmark shows Vevea-Hedges is strongest exactly when the truth is a p-step (its own form) and degrades otherwise, and it is non-identified at k<=10 (a reproducible ~1% runaway inflates its small-k RMSE). Treat the parametric selection models as k>=15 tools.

- **The general publication-selection problem is unsolved.** This study handles only

the *modelled* mechanisms (p-step and Copas). A genuinely unknown or adversarial selection process is not covered; the Unified interval targets coverage under the mixture it was trained/bounded over, not under arbitrary selection. We claim improved calibrated coverage under heterogeneity and *known* selection, and honest partial-identification bounds -- **not** a solution to publication bias.

- **NPE amortization assumptions.** The amortized NPE is calibrated for the training

prior's mixture of mechanisms and severities and the observable-feature regime; a meta-analysis far outside that support (e.g. extreme k, effect scales, or a mechanism absent from the training mixture) is outside its guarantee. The conformal layer targets finite-sample coverage *within* that support.

- **Effect-scale and dependence scope.** Studies are scalar effects with known within-

study variances and are independent; correlated/multi-arm effects, network structure, and unknown within-study variance are out of scope (see the NMA and DTA modalities of this bench for the dependent cases).

- **Coverage is simulation-based.** We make no claim of measured coverage on real

data, where the truth is unknown; the real-data claim is *correctness* against the R references (§4), not coverage.

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6. Reproducibility statement

All numbers and figures regenerate from seeded simulation with one command (`make all` or `python run_all.py`). Determinism: each replication draws from `np.random.default_rng(SeedSequence([20260611, stable_hash(cell_id), k]).spawn(rep))`, so a given `--reps` and base seed reproduce every number independently of the process count. Pinned core dependencies are in `requirements.txt` (numpy 2.4.4, scipy 1.17.1, matplotlib 3.10.9, reportlab 4.5.1, pytest 9.0.3, scikit-learn 1.8.0 on Python 3.13). The committed `results_merged.json` is the exact output that produced §3 (the incumbent run `results_full.json` plus the grafted NPE/PVS/PartialID/Unified metrics); `REPORT.md` is its auto-generated tabular summary.

The amortized NPE model `sbi_model.pkl` is a **committed, pre-computed artifact**; `run_all.py` loads it to score NPE/Unified and does **not** retrain it. Retraining is a separate, documented, offline step (`python train_sbi.py`) needing only scikit-learn (this benchmark's SBI estimator uses scikit-learn's `HistGradientBoostingRegressor`, not torch or the `sbi` PyPI package; see `requirements-train.txt` and `DATA_MANIFEST.md`). The `--no-sim` flag rebuilds figures, the worked example, the PDF and the test gate from the committed JSON in seconds.

Code and data availability. Source, tests, results JSON, the pre-trained model, calibration diagnostics, figures and this manuscript are in the `truth-recovery-integration` branch of the truth-recovery-bench repository. The simulation study requires no external data; the Copas correctness anchor reads the audited `copas-oracle.json` fixture from the sibling allmeta Copas parity suite.

References (indicative)

1. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Control Clin Trials*

1986.

2. Hartung J, Knapp G. A refined method for the meta-analysis of controlled clinical

trials. *Stat Med* 2001. Sidik K, Jonkman JN. 2002.

3. IntHout J, Ioannidis JPA, Borm GF. The Hartung-Knapp-Sidik-Jonkman method...

considerably outperforms the standard DerSimonian-Laird method. *BMC Med Res Methodol* 2014.

4. Vevea JL, Hedges LV. A general linear model for estimating effect size in the

presence of publication bias. *Psychometrika* 1995.

5. Copas J, Shi JQ. Meta-analysis, funnel plots and sensitivity analysis.

Biostatistics 2000.

6. Viechtbauer W. Conducting meta-analyses in R with the metafor package. *J Stat

Softw* 2010.

7. Schwarzer G, Carpenter JR, Rucker G. metasens: Statistical methods for sensitivity

analysis in meta-analysis. R package.

8. Manski CF. *Partial Identification of Probability Distributions.* Springer 2003.

9. Romano Y, Patterson E, Candes E. Conformalized quantile regression. *NeurIPS*

2019.

10. Cranmer K, Brehmer J, Louppe G. The frontier of simulation-based inference.

PNAS 2020.